

Cheat Sheet — Ch 1 · 2 · 3 · 4 · 7

CONSTANTS $h = 6.626 \times 10^{-34} \text{ J s}$ · $c = 3 \times 10^8 \text{ m/s}$ · $e = 1.602 \times 10^{-19} \text{ C}$ · $m_e = 9.11 \times 10^{-31} \text{ kg}$ · $N_A = 6.022 \times 10^{23}$ · $R_H = 109677 \text{ cm}^{-1}$ · $E_1(\text{H}) = -13.6 \text{ eV}$ · $r_1 = 52.9 \text{ pm}$ · $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ · STP: **22.7 L @ 1 bar**, 22.4 L @ 1 atm

CH 1 · MOLE — THE FOUR BRIDGES

$$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V}{22.7\text{L}} = M \times V(L)$$

Avg atomic mass = $\Sigma(\text{abundance} \times \text{isotopic mass})$. %E = (mass of E in 1 mol / molar mass) $\times 100$.

$$\text{MF} = (\text{EF})_n, \quad n = \frac{M}{\text{EFM}}$$

Limiting reagent: moles $\div$ coefficient $\rightarrow$ **smallest wins**. Never compare masses.

$$\% \text{ yield} = \text{actual/theoretical} \times 100$$

CH 2 · PARTICLES & MODELS

$e/m = 1.7588 \times 10^{11} \text{ C/kg}$ (same for all gases $\Rightarrow$ electron universal) · $e = 1.602 \times 10^{-19} \text{ C}$ (Millikan $\Rightarrow$ charge quantised) · p 1837 $\times e^-$ · n slightly $>$ p.

Anode rays: e/m varies — they are the **ionised gas atoms**; mass and charge both change.

α -scattering: most through $\rightarrow$ empty space · few deflect $\rightarrow$ charge concentrated · 1 in 20000 rebounds $\rightarrow$ also massive. Nucleus 10^{-15} , atom 10^{-10} m .

Rutherford drawbacks (2): accelerating e^- radiates $\rightarrow$ spirals in $\sim 10^{-8} \text{ s}$; would give **continuous** not **line** spectrum.

CH 2 · SPECTRUM & BOHR

$$\bar{\nu} = 109677 \left(\frac{1}{n_1} - \frac{1}{n_2} \right) \text{ cm}^{-1}$$

Series	n_1	Region
Lyman	1	UV
Balmer	2	Visible
Paschen	3	IR
Brackett / Pfund	4 / 5	IR

Named by where e^- **lands**. Lines to ground = $n(n-1)/2$.

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm} \cdot E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} \cdot mvr = \frac{nh}{2\pi} \cdot \Delta E = h\nu$$

Limitations (5): one- e^- species only · no fine structure · no Zeeman/Stark · no bonding · violates Heisenberg.

CH 3 · TRENDS

	Across $\rightarrow$	Down $\downarrow$
Atomic radius	Decreases	Increases
Ionization enth.	Increases	Decreases
Electron gain enth.	More -ve	Less -ve
Electronegativity	Increases	Decreases
Metallic char.	Decreases	Increases
Oxide	basic $\rightarrow$ amph $\rightarrow$ acidic	more basic

Why: across — $Z \uparrow$ with **same shell** $\Rightarrow$ pull $\uparrow$. Down — **new shell** + shielding $\uparrow$ $\Rightarrow$ pull on outer e^- **weakens**.

Cation $<$ atom $<$ anion. Isoelectronic: **more Z $\Rightarrow$ smaller**.

$$\text{O}^{2-}140 > \text{F}^{-}136 > \text{Na}^{+}95 > \text{Mg}^{2+}65 > \text{Al}^{3+}50 \text{ pm}$$

CH 1 · CONCENTRATION

Term	Formula	T?
Mass %	$\frac{w_B}{w_A + w_B} \times 100$	No
Mole frac	$x_B = \frac{n_B}{n_A + n_B}$	No
Molarity	$M = \frac{n_B}{V_{\text{soln}}(L)}$	Yes
Molality	$m = \frac{n_B}{w_A(\text{kg})}$	No
ppm	$\frac{w_{\text{solute}}}{w_{\text{soln}}} \times 10^6$	No

$$M = \frac{10 \times \% \times d}{M_B} \cdot m = \frac{1000M}{1000d - M M_B} \cdot M_1 V_1 = M_2 V_2$$

Sig figs: $+/- \rightarrow$ fewest *decimals*; $\times/\div \rightarrow$ fewest *sig figs*.

CH 2 · RADIATION, PHOTON, PHOTOELECTRIC

$$c = \nu \lambda \cdot \bar{\nu} = 1/\lambda \cdot E = h\nu = \frac{hc}{\lambda} = nh\nu$$

$$h\nu = h\nu_0 + \frac{1}{2}m_e v^2 \cdot \phi = h\nu_0 \cdot KE_{\text{max}} = h(\nu - \nu_0)$$

Graph KE vs ν : straight line, **slope = h** (same all metals $\Rightarrow$ parallel), x-intercept = ν_0 . No emission below ν_0 .

4 observations: threshold ν_0 · instantaneous · $KE \propto$ frequency · number $\propto$ intensity.

CH 2 · DE BROGLIE, HEISENBERG, QM

$$\lambda = \frac{h}{mv} \cdot \Delta x \cdot \Delta p \geq \frac{h}{4\pi} \cdot \hat{H}\psi = E\psi$$

ψ no meaning; **ψ^2 = probability density**. Orbital = 90% region, max 2 e^- .

QN	Values	Fixes
n	1,2,3...	shell, size; $2n^2 e^-$
l	$0 \rightarrow n-1$	subshell, shape ; $L = \sqrt{l(l+1)}h/2\pi$
m_l	$-l \dots +l$	orientation; $2l+1$ orbitals
m_s	$\pm 1/2$	spin (not from Schrödinger)

Nodes: angular = l · radial = $n-l-1$ · **total = $n-1$** . Orbitals/shell = n^2 .

Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p. $(n+l)$ rule; tie $\rightarrow$ lower n.

Cr = [Ar]3d⁵4s¹, Cu = [Ar]3d¹⁰4s¹ — symmetry + exchange energy. **Fe²⁺ = [Ar]3d⁶** (4s empties first).

H only: 2s = 2p. All other atoms: $s < p < d < f$.

CH 3 · THE FOUR ANOMALIES

Be > B, Mg > Al (IE): B's e^- from **2p** — higher energy, better shielded — vs Be's filled 2s².

N > O, P > S (IE): N is **half-filled 2p³**, symmetric & stable; O's **paired** 2p e^- repels $\Rightarrow$ easier to remove.

Cl > F, S > O ($\Delta_{\text{eg}}H$ more -ve): F/O so **small** that compact 2p causes **inter-electronic repulsion**.

2nd $\Delta_{\text{eg}}H$ always +ve; noble gases +ve.

CH 3 · BLOCKS & DATA

s gp 1-2 ns^{1-2} · **p** gp 13-18 ns^2np^{1-6} · **d** gp 3-12 $(n-1)d^{1-10}ns^{0-2}$ · **f** $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

Superscript dash = range. (n-1) = real minus. Block width = subshell capacity.

Zn/Cd/Hg not transition (d^{10} in atom & ion). Period lengths **2,8,8,18,18,32,32**.

Group: s = valence e^- ; p = 10 + (ns+np); d = (n-1)d + ns.

IE period 2: Li 520, **Be 899, B 801**, C 1086, **N 1402, O 1314**, F 1681, Ne 2080

Δ_{eg} H: F -328, **Cl -349**, O -141, **S -200** · EN: **F = 4.0**

2nd-period anomaly (3): small size · high charge/radius & EN · **no d orbitals** $\Rightarrow$ covalency max **4**. **Diagonal:** Li-Mg, Be-Al, B-Si.

Roots >100: nil un bi tri quad pent hex sept oct enn + **ium**. 104 = unnilquadium (Unq).

CH 4 · VSEPR — GEOMETRY VS SHAPE

BP/LP	Geometry	Shape	Eg (angle)
2/0	Linear	Linear	CO ₂ 180°
3/0	Trig planar	Trig planar	BF ₃ 120°
2/1	Trig planar	Bent	SO ₂
4/0	Tetrahedral	Tetrahedral	CH ₄ 109.5°
3/1	Tetrahedral	Pyramidal	NH ₃ 107°
2/2	Tetrahedral	Bent	H ₂ O 104.5°
5/0	Trig bipy	Trig bipy	PCl ₅ 90/120°
4/1	Trig bipy	See-saw	SF ₄
3/2	Trig bipy	T-shaped	ClF ₃
2/3	Trig bipy	Linear	XeF ₂
6/0	Octahedral	Octahedral	SF ₆ 90°
5/1	Octahedral	Sq pyramidal	BrF ₅
4/2	Octahedral	Sq planar	XeF ₄

Repulsion **lp-lp > lp-bp > bp-bp**. LP goes **equatorial** in TBP, **trans** in octahedral.

Angles: CH₄109.5 > NH₃107 > H₂O104.5 · NH₃ > NF₃102 (EN of F) · NH₃ > PH₃93.6 (size).

CH 4 · POLARITY & H-BONDING

$\mu = 0$: BeF₂, CO₂, BF₃, CCl₄ (symmetric). H₂O 1.85, **NH₃ 1.47** > **NF₃ 0.24**.

NH₃ vs NF₃: in NH₃ bond dipoles point **toward N**, **adding** to the lone-pair dipole; in NF₃ they point **away**, **opposing** it.

H-bond needs H on **F, O or N**. 10-40 kJ/mol. Inter $\rightarrow$ **raises** bp; intra (o-nitrophenol) $\rightarrow$ **lowers** bp. Ice open cage = less dense.

Resonance: only e^- move, nuclei fixed; forms not real; hybrid lower in energy; O₃ both bonds 128 pm.

CH 4 · CORE FORMULAE

$FC = V - L - \frac{1}{2}B$ · $H = \frac{1}{2}[V + M - C + A]$ · $BO = \frac{N_A - N_B}{2}$ · $\mu = q \times d$

H = 2 sp · 3 sp² · 4 sp³ · 5 sp³d · 6 sp³d². O/S divalent $\rightarrow$ contribute 0 to M.

Octet fails: incomplete (BCl₃) · expanded (SF₆) · odd- e^- (NO, NO₂) · noble gas cpds.

CH 4 · MOT

$\leq 14 e^-$ (B₂, C₂, N₂): $\sigma 2s < \sigma^* 2s < \pi 2p = \pi 2p < \sigma 2p < \pi^* < \sigma^*$
> 14 (O₂, F₂): $\sigma 2p < \pi 2p$ (swap)

	H ₂	He ₂	B ₂	C ₂	N ₂	O ₂	F ₂
BO	1	0	1	2	3	2	1
Mag	dia	—	para	dia	dia	para	dia

O₂⁺ 2.5 > O₂ 2 > O₂⁻ 1.5 > O₂²⁻ 1 (bond length reverse; only O₂²⁻ diamagnetic).

σ head-on, stronger, free rotation, always 1st. **π** sideways, weaker, restricted. Double = 1 σ +1 π ; triple = 1 σ +2 π .

CH 7 · OXIDATION NUMBERS

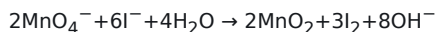
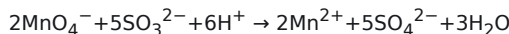
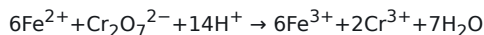
Free element **0** · monatomic ion = charge · **F always -1** · gp1 +1, gp2 +2 · H +1 but **-1 in hydrides** · O -2, **-1 peroxide**, **-1/2 superoxide**, **+ve with F** · sum = 0 or charge.

Cr in Cr₂O₇²⁻ +6 · Mn in MnO₄⁻ +7 · **Fe₃O₄ +8/3** · **S₄O₆²⁻ +2.5** (averages).

ON can be $\pm$, 0, fractional. **Valency cannot**. OIL RIG. Oxidising agent is **reduced**.

CH 7 · BALANCING (HALF-REACTION)

1. Split into oxidation & reduction halves
2. Balance atoms except H, O
3. Balance **O with H₂O**
4. Balance **H with H⁺**
5. Balance **charge with e⁻**
6. Multiply so e⁻ cancel; add
7. **Basic:** add OH⁻ to **both** sides = no. of H⁺; combine to H₂O



Always check total charge, not just atoms.

CH 7 · TITRATION & CELLS

Medium	MnO ₄ ⁻ →	n
Acidic	Mn ²⁺ + 4H ₂ O	5
Neutral	MnO ₂ + 4OH ⁻	3
Strong alkali	MnO ₄ ²⁻	1
Cr ₂ O ₇ ²⁻ acid	2Cr ³⁺ + 7H ₂ O	6

$E = \frac{M}{n} \cdot N = M \times n \cdot N_1 V_1 = N_2 V_2 \cdot E_{cell}^{\circ} = E_{cat}^{\circ} - E_{an}^{\circ}$ (+ve ⇒ spontaneous)

Use dil. H₂SO₄ only — HCl oxidised to Cl₂, HNO₃ is itself an oxidant.

Disproportionation needs an **intermediate** ON.

2H₂O₂ → 2H₂O + O₂; Cl₂ + 2OH⁻; 2Cu⁺ → Cu²⁺ + Cu. **F cannot** (no +ve state).

Daniell: Zn|Zn²⁺||Cu²⁺|Cu, E° = +1.10 V. Anode = -ve = oxidation.

FINAL LOOK — THE TRAPS

1. STP **22.7 L at 1 bar**, 22.4 at 1 atm.
2. Molarity T-dependent; molality not.
3. Limiting reagent: moles ÷ coefficient.
4. Photoelectric KE → **frequency**; current → intensity.
5. Bohr: **one-electron species only**.
6. Series named by **landing** level.
7. Total nodes = n - 1. s orbital L = **0**.
8. **l never = n** (no 2d, 1p, 3f).
9. 4s fills first, **empties first**.
10. Isoelectronic: more Z = smaller.
11. **Geometry ≠ shape** (H₂O bent; XeF₄ sq planar).
12. O₂ paramagnetic — **MOT only**.
13. Adding e⁻ to O₂ **weakens** it (antibonding).
14. Basic medium → finish with **OH⁻**.
15. E° values are **never** multiplied by coefficients.